Tutorial-7

MID	AS	RS	G
1	15	2	A
2	1	12	R
3	18	1	A
4	2	14	R
5	14	3	A

Now: (10, 5.) validation time! pick single movie id that is least 4.47, Action

$D_1 = \sqrt{(10-x)^2+(5-y)^2}$

$= \sqrt{(10-15)^2+(5-2)^2}$

$= \sqrt{(5)^2+(3)^2}$

$= \sqrt{25+9}$

$= \sqrt{34} = \underline{5.83}$

$D_2 = \sqrt{(10-1)^2+(5-12)^2}$

$= \sqrt{81+49} = \underline{11.40}$

$D_3 = \sqrt{(10-18)^2+(5-1)^2}$

$= \sqrt{64+16}$

$= \underline{8.94}$

$D_4 = \sqrt{(10-2)^2+(5-14)^2}$

$= \sqrt{64+81} = \underline{12.04}$

$D_5 = \sqrt{(10-14)^2+(5-3)^2}$

$= \sqrt{16+4} = \sqrt{20}$

$= \underline{4.47}$

$D_6 = \sqrt{(10-3)^2+(5-11)^2}$

$= \sqrt{49+36} = \underline{9.21}$

$D_7 = \sqrt{(10-16)^2+(5-2)^2}$

$= \sqrt{36+9} = \underline{6.70}$

$D_8 = \sqrt{(10-2)^2+(5-13)^2}$

$= \sqrt{64+64} = 11.31$

K_3 = least three

id = 5, 1, 7

4.47, 5.38, 6.70 } Action

MID	Dist	Genre
5	4.47	A
1	5.38	
7	6.70	
3	8.94	
6	9.21	
8	11.31	
2	11.40	
4	12.04	

$T_2(9,10)$

$D_1 = \sqrt{(9-x)^2+(10-y)^2}$

$= \sqrt{(9-15)^2+(10-2)^2}$

$= \sqrt{121+69} = \underline{13.60}$

$D_2 = \sqrt{(9-1)^2+(10-11)^2}$

$= \sqrt{9+9} = \underline{3.60}$

$D_3 = \sqrt{(9-18)^2+(10-1)^2}$

$= \sqrt{(14)^2+(9)^2} = \sqrt{277} = \underline{16.64}$

$D_4 = \sqrt{(9-2)^2+(10-14)^2}$

$= \sqrt{4+16}$

$= \sqrt{20} = \underline{4.47}$

$D_5 = \sqrt{(9-15)^2+(10-3)^2}$

$= \sqrt{121+49} = \underline{13.03}$

$D_6 = \sqrt{(9-3)^2+(10-11)^2}$

$= \sqrt{1+1} = \sqrt{2} = \underline{1.41}$

$D_7 = \sqrt{(9-16)^2+(10-2)^2}$

$= \sqrt{144+64} = \sqrt{208} = \underline{14.42}$

$D_8 = \sqrt{(9-2)^2+(10-13)^2}$

$= \sqrt{4+9}$

$= \sqrt{13} = \underline{3.60}$

$\underline{K3} \Rightarrow$ 6 ID - 1.41
2 ID = 3.60
6 ID = 3.60 } R

$T_3(17,2)$

$D_1 = \sqrt{(17-15)^2+(2-2)^2}$

$= \sqrt{4+0} = \sqrt{4} =$

$D_2 = \sqrt{(17-1)^2+(2-11)^2}$

$= \sqrt{256+100} = 18.$

$D_3 = \sqrt{(17-18)^2+(2-1)^2}$

$= \sqrt{1+1} = \sqrt{2} = 1$

$D_4 = \sqrt{(17-2)^2+(2-14)^2}$

$= \sqrt{(225+144)}$

$= 19.20$

$D_5 = \sqrt{(17-15)^2+(2-3)^2}$

$= \sqrt{4+1} = \sqrt{5}$

$= 2.23$

$D_6 = \sqrt{(17-3)^2+(2-11)^2}$

$= \sqrt{277} = 16.64$

$D_7 = \sqrt{(17-16)^2+(2-2)^2}$

$= \sqrt{1+0} = \sqrt{1} =$

$D_8 = \sqrt{(17-2)^2+(2-13)^2}$

$= \sqrt{225+121}$

$= \sqrt{346}$

$= 18.60$

$K_5 \rightarrow$ least 5

$D_1 \rightarrow 1 = A$

$D_3 \rightarrow 1.41 = A$

$D_1 \rightarrow 2 = A$

$D_5 \rightarrow 2.23 = A$

$D_6 \rightarrow$

Confusion Matrix

Actual / Predicted →	A	R
A	2	0
R	0	1

$= \frac{0.875}{5}$

$= 0.175$

$MAE = \frac{1}{N}\sum_{i=1}^{N} |y_i - \bar{y}|$

$= 0.6$

$1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y})^2}{\sum^{N} (y_i - \bar{y})^2}$

$= \frac{0.125}{0.5}$

$MSE = \frac{1}{N}\sum_{i=1}^{N} (y_i - \bar{y})^2$

13/03/26